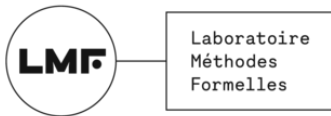


Elements of Higher-Dimensional Automata Theory

Uli Fahrenberg

LMF, Université Paris-Saclay, France

LAP Invited Talk, September 2026



Language theory of higher-dimensional automata

- Languages of Higher-Dimensional Automata [MSCS 2021]
- Kleene Theorem for Higher-Dimensional Automata [LMCS 2024]
- Myhill-Nerode Theorem for Higher-Dimensional Automata [FI 2024]
- Decision and Closure Properties for Higher-Dimensional Automata [TCS 2025]
- Logic and Languages of Higher-Dimensional Automata [DLT 2024]
- Higher-Dimensional Automata: Extension to Infinite Tracks. [FSCD 2025]
- Kamp Theorem for Higher Dimensional Automata. [CSL 2026]
- Forgetting Event Order in Higher-Dimensional Automata. [LICS 2026]
- Algebraic Characterization of FO-definable Languages of HDAs. [CONCUR 2026]
- Bisimulations and Modal Logics for HDAs. [CONCUR 2026]

Today:

- ① What are HDAs?
- ② What are languages of HDAs?
- ③ What can I do with languages of HDAs?

Nice people

Amazigh **Amrane**, Hugo **Bazille**, Paul **Brunet**, Emily **Clement**, Jérémy **Dubut**, Marie **Fortin**, Eric **Goubault**, Emmanuel **Haucourt**, Loïc **Hélouët**, Christian **Johansen**, Jérémy **Ledent**, Samuel **Mimram**, Philipp **Schlehuber-Caissier**, Georg **Struth**, Rob **van Glabbeek**, Krzysztof **Ziemiański**, ...

See also <https://ulifahrenberg.github.io/pomsetproject/>

CHAAT

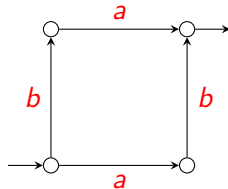


- French national research project (**ANR**), starting October 2026
 - **Concurrency and Higher-dimensional Automata: Applications and Theory**
 - LMF, LRE, IRIF, LIX, SAMOVAR
- Logo by Paul Brunet (LACL)

- 1 Introduction
- 2 Higher-Dimensional Automata
- 3 Languages of Higher-Dimensional Automata
- 4 Properties
- 5 Kleene theorem
- 6 Büchi-Elgot-Trakhtenbrot theorem
- 7 Conclusion

Higher-dimensional automata

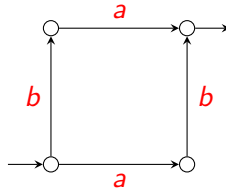
semantics of “a parallel b”:



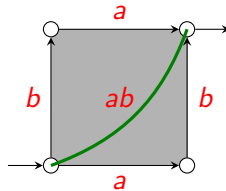
$$a.b + b.a$$

Higher-dimensional automata

semantics of “a parallel b”:



$$a.b + b.a$$



a and b are independent

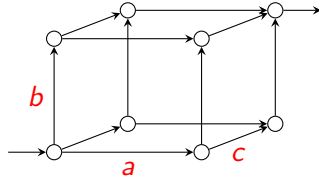
Higher-dimensional automata & concurrency

HDAs as a model for **concurrency**:

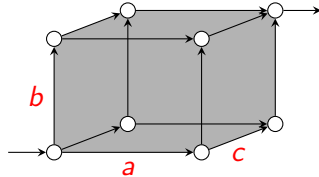
- points: **states**
- edges: **transitions**
- squares, cubes etc.: **independency** relations / **concurrently** executing events
- **two**-dimensional automata $\cong$ asynchronous transition systems [Bednarczyk]

[van Glabbeek 2006, TCS]: Up to history-preserving bisimilarity, HDAs “generalize the main models of concurrency proposed in the literature” (notably, event structures and Petri nets)

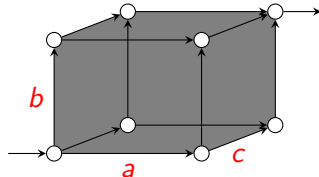
Examples



no concurrency



two out of three



full concurrency

Higher-dimensional automata

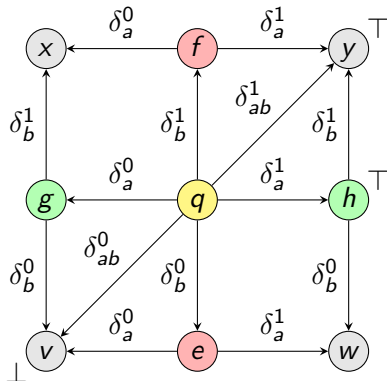
A **conclist** is a finite, ordered and Σ -labelled set. (a list of events)

A **precubical set** X consists of:

- A set of cells X (cubes)
- Every cell $x \in X$ has a conclist $\text{ev}(x)$ (list of events active in x)
- We write $X[U] = \{x \in X \mid \text{ev}(x) = U\}$ for a conclist U (cells of type U)
- For every conclist U and $A \subseteq U$ there are:
 - upper face map $\delta_A^1 : X[U] \rightarrow X[U \setminus A]$ (terminating events A)
 - lower face map $\delta_A^0 : X[U] \rightarrow X[U \setminus A]$ (“unstarting” events A)
- **Precube identities:** $\delta_A^\mu \delta_B^\nu = \delta_B^\nu \delta_A^\mu$ for $A \cap B = \emptyset$ and $\mu, \nu \in \{0, 1\}$

A **higher dimensional automaton (HDA)** is a precubical set X with **start cells** $\perp \subseteq X$ and **accept cells** $\top \subseteq X$ (not necessarily vertices)

Example



$$X[\emptyset] = \{v, w, x, y\}$$

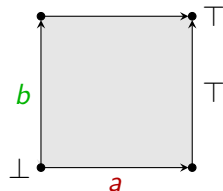
$$X[a] = \{e, f\}$$

$$X[b] = \{g, h\}$$

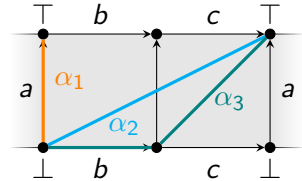
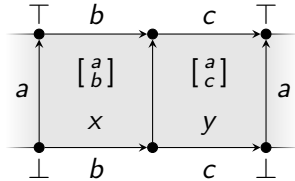
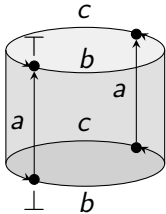
$$X[ab] = \{q\}$$

$$\perp_X = \{v\}$$

$$\top_X = \{h, y\}$$



Another one

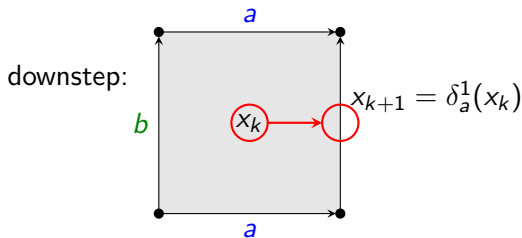
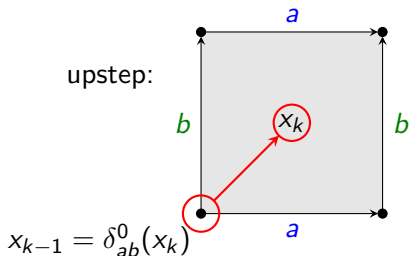


$$a \parallel (bc)^*$$

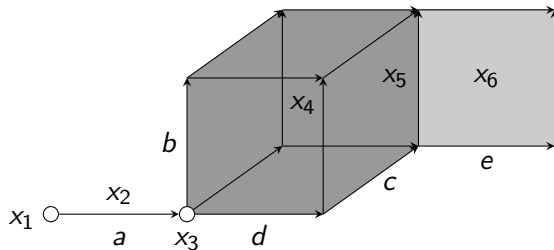
Computations of HDAs

A **path** on an HDA X is a sequence $(x_0, \varphi_1, x_1, \dots, x_{n-1}, \varphi_n, x_n)$ such that for every k , $(x_{k-1}, \varphi_k, x_k)$ is either

- $(\delta_A^0(x_k), \nearrow^A, x_k)$ for $A \subseteq \text{ev}(x_k)$ or (upstep: start A)
- $(x_{k-1}, \searrow_B, \delta_B^1(x_{k-1}))$ for $B \subseteq \text{ev}(x_{k-1})$ (downstep: terminate B)



Example



$$(x_1 \nearrow^a x_2 \searrow_a x_3 \nearrow^{\{b,c,d\}} x_4 \searrow_{\{c,d\}} x_5 \nearrow^e x_6)$$

Presheaves

Definition (van Glabbeek, 1989)

A **precubical set** is a graded set $X = \{X_n\}_{n \geq 0}$ together with **face maps** $\delta_i^0, \delta_i^1 : X_n \rightarrow X_{n-1}$, for $i = 1, \dots, n$, satisfying $\delta_i^\nu \delta_j^\mu = \delta_{j-1}^\mu \delta_i^\nu$ for $i < j$.

- presheaves over a small **precube category** $\square$

Definition (us, 2020–)

A **precubical set** is a set X together with a mapping $\text{ev} : X \rightarrow (\text{conclists})$ and with **face maps** $\delta_A^0, \delta_A^1 : X[U] \rightarrow X[U \setminus A]$ satisfying $\delta_A^\mu \delta_B^\nu = \delta_B^\nu \delta_A^\mu$ for $A \cap B = \emptyset$.

- presheaves over another, **large** precube category $\square^\circ$
- natural notion of **events**

Presheaves Presheaves Presheaves

(augmented) precube category $\square$	large (augmented) precube category $\square^\bullet$
objects $\{0, 1\}^n$ for $n \geq 0$	objects totally ordered finite sets
morphisms injections of 0s and 1s	morphisms A, B -injections
skeletal	isos are unique

Lemma

The inclusion $\square \hookrightarrow \square^\bullet$ is an **equivalence** of categories with a **unique** left inverse.

Corollary

The presheaf categories $\text{Set}^{\square^{\text{op}}}$ and $\text{Set}^{\square^{\bullet\text{op}}}$ are **uniquely naturally isomorphic**.

- **precubical sets**: $\text{Set}^{\square^{\text{op}}}$ or $\text{Set}^{\square^{\bullet\text{op}}}$; makes no difference

Presheaves Presheaves Presheaves

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- **precubical sets**: $\text{Set}^{\square^{\text{op}}}$ or $\text{Set}^{\square^{\bullet \text{op}}}$; makes no difference
- (but, no labels in $\square^\bullet$; **fundamental theorem** takes care of this:
For $\mathcal{C}$ a presheaf cat and $\Sigma \in \mathcal{C}$, also the slice $\mathcal{C}/\Sigma$ is a presheaf cat.)

Context

(augmented) precube category $\square$	large (augmented) precube category $\square^\bullet$
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$\square \hookrightarrow \square^\bullet$ equivalence with unique left inverse	

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(augmented) precube category $\square$ objects $\{0, 1\}^n$ for $n \geq 0$ morphisms injections of 0s and 1s skeletal	large (augmented) precube category $\square$ objects totally ordered finite sets morphisms A, B-injections isos are unique
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augmented presimplex category Δ objects $\{1 < \dots < n\}$ for $n \geq 0$ morphisms order injections skeletal	large augmented presimplex category Δ objects totally ordered finite sets morphisms order injections isos are unique
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Context

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	category of species $\mathbb{B}$ objects finite sets morphisms bijections isos are not unique

Context

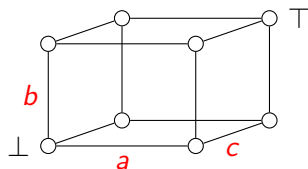
(augmented) precube category $\square$ objects $\{0, 1\}^n$ for $n \geq 0$ morphisms injections of 0s and 1s skeletal	large (augmented) precube category $\square$ objects totally ordered finite sets morphisms A, B-injections isos are unique
$\square \hookrightarrow \square$ equivalence with unique left inverse	
augmented presimplex category Δ objects $\{1 < \dots < n\}$ for $n \geq 0$ morphisms order injections skeletal	large augmented presimplex category Δ objects totally ordered finite sets morphisms order injections isos are unique
$\Delta \hookrightarrow \Delta$ equivalence with unique left inverse	
“category” of ordinals objects $\{1, \dots, n\}$ for $n \geq 0$ morphisms permutations skeletal	category of species $\mathbb{B}$ objects finite sets morphisms bijections isos are not unique

The Zoo of Cubes (It's HoTT!)

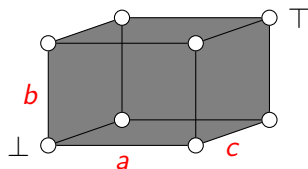
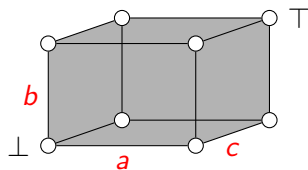
precubical set: graded set $X = \{X_n\}_{n \geq 0}$ plus **face maps** $\delta_i^0, \delta_i^1 : X_n \rightarrow X_{n-1}$

- plus **degeneracies** $\epsilon_i : X_n \rightarrow X_{n+1}$: **cubical** set
- with **connections** $\gamma_i^0, \gamma_i^1 : X_n \rightarrow X_{n+1}$
- with **transpositions** $\sigma_i : X_n \rightarrow X_n$
- with **diagonals** $\Delta : X_n \rightarrow X_{n-1}$
- many subsets of these are in use
- all are presheaves
- diagonals are important in **cubical homotopy type theory**
- cubical **ω -categories** with connections are equivalent to **globular** ω -categories

Languages of HDAs: Examples

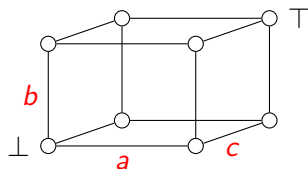


$$L_1 = \{abc, acb, bac, bca, cab, cba\}$$

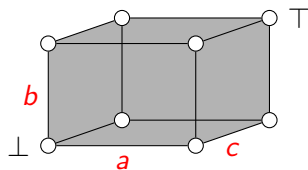


$$L_3 = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \dots \right\}$$

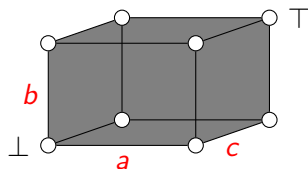
Languages of HDAs: Examples



$$L_1 = \{abc, acb, bac, bca, cab, cba\}$$



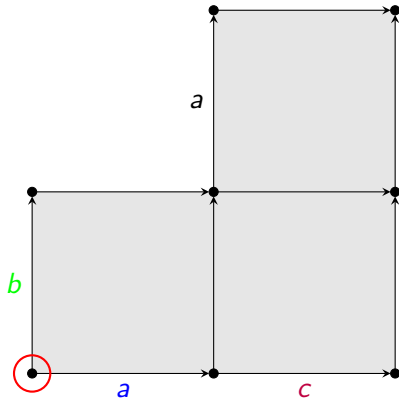
$$L_2 = \left\{ \begin{bmatrix} a \\ b \rightarrow c \end{bmatrix}, \begin{bmatrix} a \\ c \rightarrow b \end{bmatrix}, \begin{bmatrix} b \\ a \rightarrow c \end{bmatrix}, \begin{bmatrix} b \\ c \rightarrow a \end{bmatrix}, \begin{bmatrix} c \\ a \rightarrow b \end{bmatrix}, \begin{bmatrix} c \\ b \rightarrow a \end{bmatrix} \right\} \cup L_1 \cup \dots$$



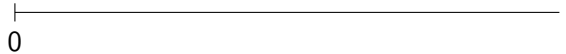
$$L_3 = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} \right\} \cup L_2$$

sets of pomsets

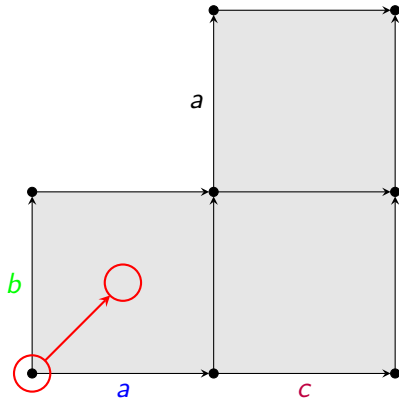
Event ipomset of a path



Lifetimes of events



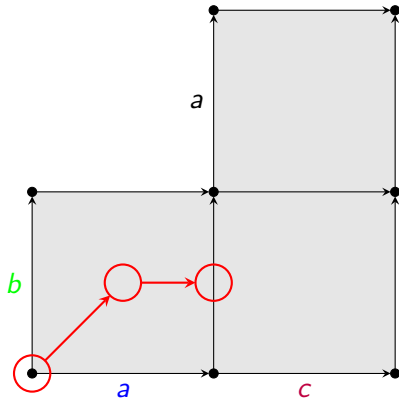
Event ipomset of a path



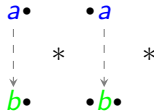
Lifetimes of events



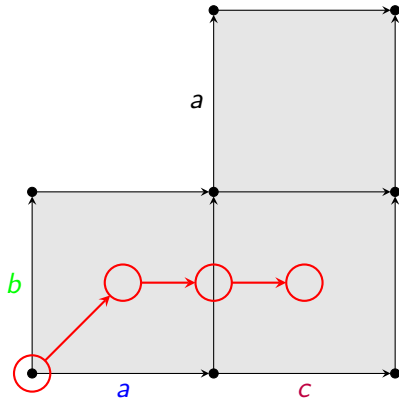
Event ipomset of a path



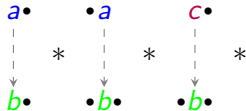
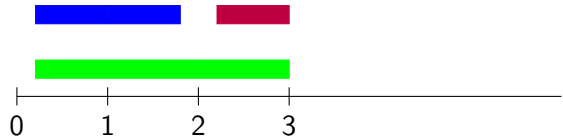
Lifetimes of events



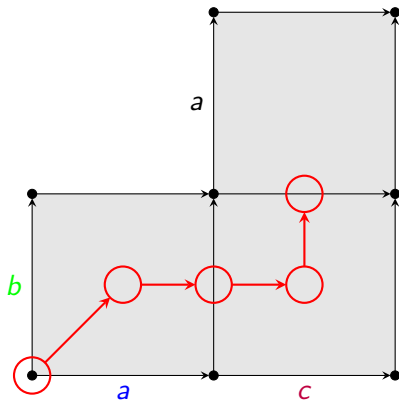
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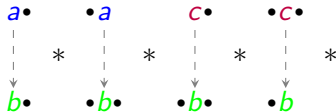
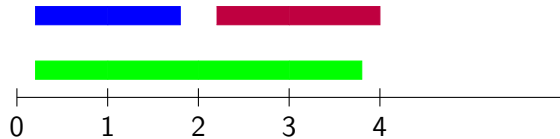
Lifetimes of events



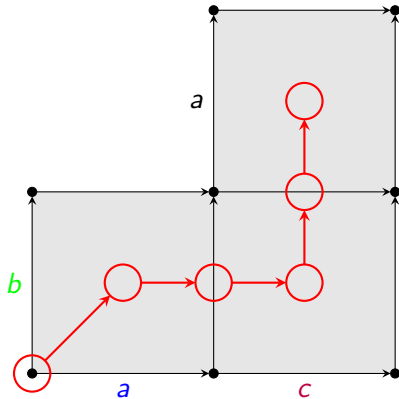
Event ipomset of a path



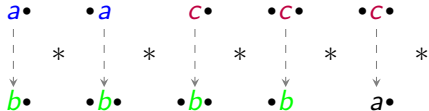
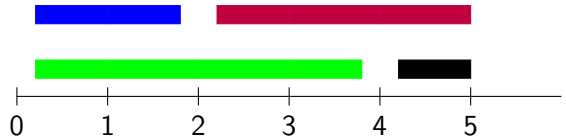
Lifetimes of events



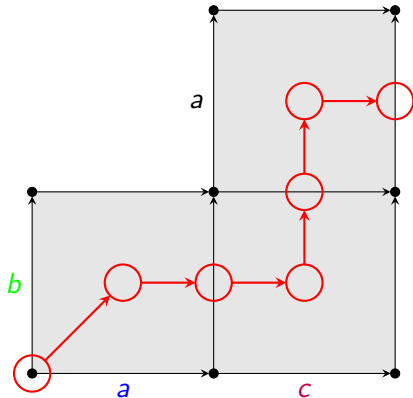
Event ipomset of a path



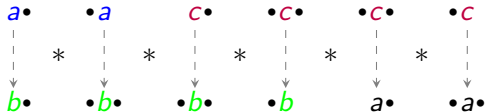
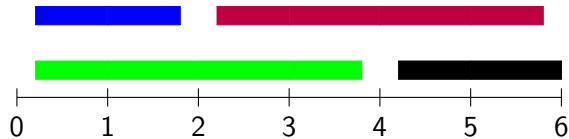
Lifetimes of events



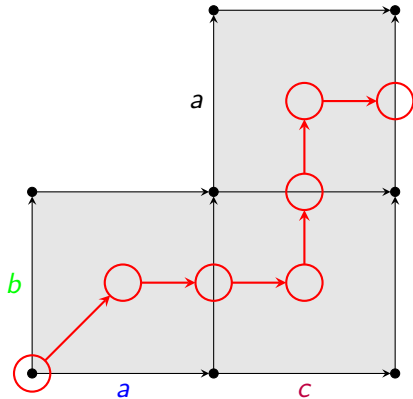
Event ipomset of a path



Lifetimes of events

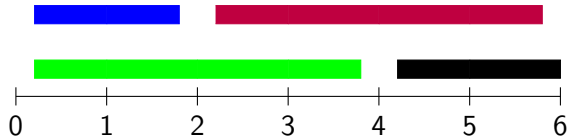


Event ipomset of a path

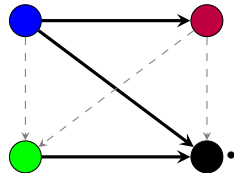


(not series-parallel!)

Lifetimes of events



Event ipomset

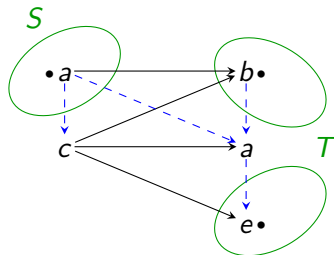


Pomsets with interfaces

Definition

A **pomset with interfaces** (**ipomset**): $(P, <, \dashrightarrow, S, T, \lambda)$:

- finite set P ;
- two partial orders $<$ (**precedence order**), $\dashrightarrow$ (**event order**)
 - s.t. $< \cup \dashrightarrow$ is a *total relation*;
- $S, T \subseteq P$ **source** and **target interfaces**
 - s.t. S is $<$ -minimal and T is $<$ -maximal.

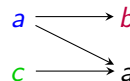
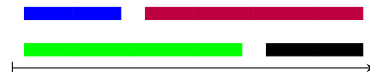
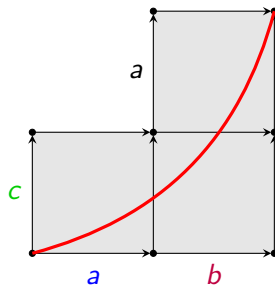


Interval orders

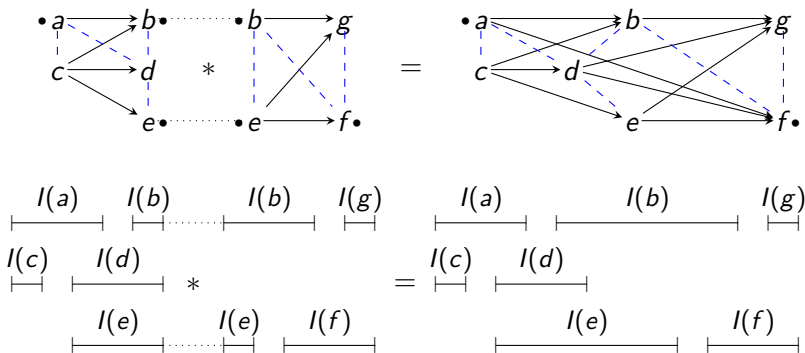
Definition

An ipomset $(P, <_P, \dashrightarrow, S, T, \lambda)$ is **interval** if $(P, <_P)$ has an **interval representation**: functions $b, e : P \rightarrow \mathbb{R}$ s.t.

- $\forall x \in P : b(x) \leq_{\mathbb{R}} e(x)$;
- $\forall x, y \in P : e(x) <_{\mathbb{R}} b(y) \iff x <_P y$

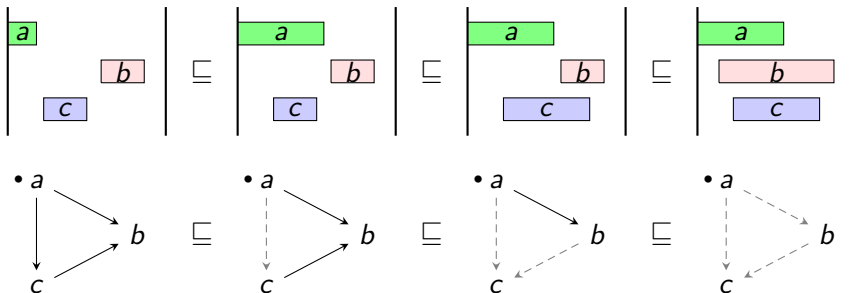


Gluing composition



- **Gluing** $P * Q$: P before Q , except for interfaces (which are identified)
- (also have **parallel composition** $P \parallel Q$: disjoint union)

Subsumption



P refines Q / Q subsumes P / $P \sqsubseteq Q$ iff

- P and Q have same interfaces
- P has more $\rightarrow$ than Q
- Q has more $\dashrightarrow$ than P

Languages of HDAs

Definition

The **language** of an HDA X is the set of event ipomsets of all accepting paths:

$$L(X) = \{\text{ev}(\pi) \mid \pi \in \text{Paths}(X), \text{src}(\pi) \in \perp_X, \text{tgt}(\pi) \in \top_X\}$$

- $L(X)$ contains only **interval** ipomsets,
- is **closed under subsumption**,
- and has **finite width**

Definition

A language $L \subseteq \text{iiPoms}$ is **regular** if there is an HDA X with $L = L(X)$.

Theorems

Definition (Rational Languages over Σ)

- Generated by $\emptyset$, $\{\epsilon\}$, and all $\{[a]\}$, $\{[\bullet a]\}$, $\{[a \bullet]\}$, $\{[\bullet a \bullet]\}$ for $a \in \Sigma$
- under operations $\cup$, $*$, $\parallel$ and (Kleene plus) $^+$
- (these need to take **subsumption closure** into account)

Definition (Monadic Second-Order Logics over Ipomsets)

$$\psi ::= a(x) \mid s(x) \mid t(x) \mid x < y \mid x \dashrightarrow y \mid x \in X \mid$$

$$\exists x. \psi \mid \forall x. \psi \mid \exists X. \psi \mid \forall X. \psi \mid \psi_1 \wedge \psi_2 \mid \psi_1 \vee \psi_2 \mid \neg \psi$$

Theorem (à la Kleene [LMCS 2024])

A language is **rational** iff it is **regular**.

Theorem (à la Büchi-Elgot-Trakhtenbrot [DLT 2024])

A language is **rational** iff it is **MSO-definable**, of finite width, and subsumption-closed.

More theorems

Theorem (à la Myhill-Nerode [FI 2024])

A language is *rational* iff it has finite *prefix quotient*.

Proof: $\implies$ as usual

$\impliedby$: usual MN congruence, but taking terminating interfaces into account

Theorem (Closure properties [TCS 2025])

Rational languages are *closed* under *intersection*

– but *not* under *complement*

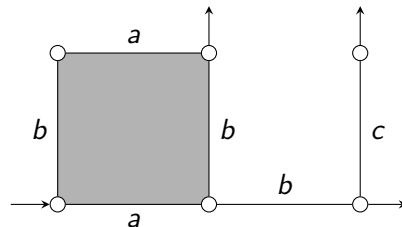
- because $L = L\downarrow$ implies $\bar{L} = \bar{L}\uparrow$; also $\bar{L}$ has *infinite width*
- *bounded-width* pseudo-complement $\bar{L}^k = \{P \in \bar{L} \mid \text{wid}(P) \leq k\}$ is useful

Determinizability & ambiguity

FI 2024

Proposition

Not all HDA's are determinizable.



Proposition

There is a rational language which is inherently infinitely ambiguous.

$$([\begin{smallmatrix} a \\ b \end{smallmatrix}] cd + ab [\begin{smallmatrix} c \\ d \end{smallmatrix}])^+$$

- ① Introduction
- ② Higher-Dimensional Automata
- ③ Languages of Higher-Dimensional Automata
- ④ Properties
- ⑤ Kleene theorem
- ⑥ Büchi-Elgot-Trakhtenbrot theorem
- ⑦ Conclusion

Kleene theorem: Easy parts

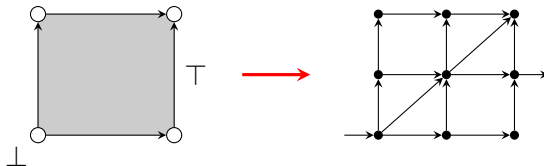
Theorem

A language is *regular* iff it is *rational*:

- generated by $\emptyset$, $\{\epsilon\}$, and all $\{[a]\}$, $\{[\cdot a]\}$, $\{[a \cdot]\}$, $\{[\cdot a \cdot]\}$ for $a \in \Sigma$
- under operations $\cup$, $*$, $\parallel$ and (Kleene plus) $^+$
- (taking subsumption closure into account)

Proof:

- regular $\implies$ rational: by reduction to **ST-automata**



Kleene theorem: Easy parts

- rational $\implies$ regular: generators:

$L(X)$	$\emptyset$	$\{\epsilon\}$	$\{[a]\}$	$\{[\bullet a]\}$	$\{[a \bullet]\}$	$\{[\bullet a \bullet]\}$
X	$\emptyset$	$\perp \circ \top$	$\begin{array}{c} \circ \top \\ \\ \perp \circ \end{array} a$	$\begin{array}{c} \circ \top \\ \\ \perp \circ \end{array} a$	$\begin{array}{c} \circ \\ \\ \perp \circ \end{array} a \top$	$\begin{array}{c} \circ \\ \\ \perp a \circ \end{array} \top$

- rational $\implies$ regular: $\cup$ and $\parallel$

$$L(X) \cup L(Y) = L(X \sqcup Y)$$

$$L(X) \parallel L(Y) = L(X \otimes Y)$$

Kleene theorem: Difficult parts

- miss to see: gluings and iterations of regular languages are regular:

$$L(X) * L(Y) = L(X * Y) \qquad L(X)^+ = L(X^+)$$

- much more difficult: higher-dimensional gluings identify too much

for example:

$L_2 = \emptyset$
 $L_1 * L_2 \ni ac$

Kleene theorem: Difficult parts

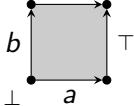
- miss to see: gluings and iterations of regular languages are regular:

$$L(X) * L(Y) = L(X * Y)$$

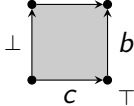
$$L(X)^+ = L(X^+)$$

- much more difficult: higher-dimensional gluings identify too much

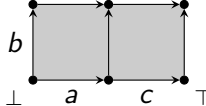
• for example:



*



=



$L_2 = \emptyset$

$L_1 * L_2 \ni ac$

- use **HDA's with interfaces** and **cylinder objects**

HDAs with interfaces

A conclist with interfaces (**iconclist**) is a conclist U with subsets $S \subseteq U \supseteq T$, denoted ${}_S U_T$
 (events in T cannot be terminated; events in S cannot be “unstarted”)

A precubical set with interfaces (**ipc-set**) X consists of a set of cells X such that:

- Every cell $x \in X$ has an **iconclist** $\text{ev}(x)$
- We write $X[{}_S U_T] = \{x \in X \mid \text{ev}(x) = {}_S U_T\}$.
- For every $A \subseteq U - S$ there is a lower face map $\delta_A^0 : X[U] \rightarrow X[{}_S U_T - A]$.
- For every $B \subseteq U - T$ there is an upper face map $\delta_B^1 : X[U] \rightarrow X[{}_S U_T - B]$.
- Precubical identities: $\delta_A^\mu \delta_B^\nu = \delta_B^\nu \delta_A^\mu$ for $A \cap B = \emptyset$ and $\mu, \nu \in \{0, 1\}$
- (presheaves over a category **I**□)

An HDA with interfaces (**iHDA**) is a finite ipc-set with start and accept cells.

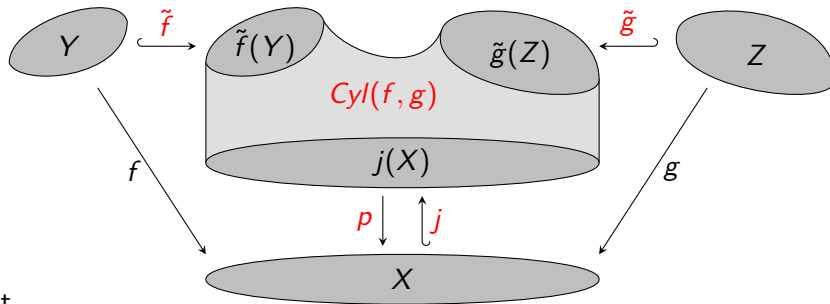
Extra conditions:

If $x \in X[{}_S U_T]$ is a start cell, then $S = U$.

If $x \in X[{}_S U_T]$ is an accept cell, then $T = U$.

Cylinders

Let X, Y, Z be ipc-sets and $f : Y \rightarrow X, g : Z \rightarrow X$ ipc-maps with $f(Y) \cap g(Z) = \emptyset$
 There is a diagram of ipc-sets



such that

- $\tilde{f}$ is an **initial inclusion**;
- $\tilde{g}$ is a **final inclusion**;
- all paths in X from $f(Y)$ to $g(Z)$ **lift** to paths in $Cyl(f, g)$.

Cylinders: construction

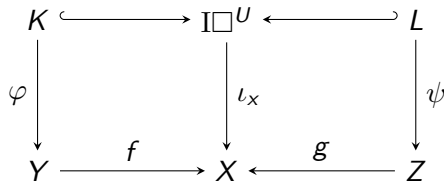
X, Y, Z : ipc-sets, $f : Y \rightarrow X, g : Z \rightarrow X$: ipc-maps with $f(Y) \cap g(Z) = \emptyset$.

For ${}_S U_T \in \mathbf{I}\Box$ let

$$\text{Cyl}(f, g)[{}_S U_T] = \{(x, K, L, \varphi, \psi)\}$$

such that

- $x \in X[{}_S U_T]$;
- $K \subseteq \mathbf{I}\Box^U$ is an initial subset;
- $L \subseteq \mathbf{I}\Box^U$ is a final subset;
- $\varphi : K \rightarrow Y, \psi : L \rightarrow Z$ are ipc-maps satisfying $f \circ \varphi = \iota_x|_K$ and $g \circ \psi = \iota_x|_L$:



Gluing composition of regular languages is regular

Proposition

Gluing composition of regular languages is regular.

Proof sketch: Let L and M be regular languages.

- ① We may assume that L, M are **simple**, i.e., $L = L(X)$, $M = L(Y)$ for iHDAs X, Y having **one initial** and **one accepting cell** each.
- ② Now replace X by $X' = \text{Cyl}(X \leftarrow \top_X : j)$ and Y by $Y' = \text{Cyl}(i : \perp_Y \rightarrow Y)$, then $L(X') = L(X)$ and $L(Y') = L(Y)$.
- ③ Go back to HDA and glue:

$$L(\mathbf{CI}(X') * \mathbf{CI}(Y')) = L(X') * L(Y') = L * M.$$

(**closure** $\mathbf{CI} : \text{iHDA} \rightarrow \text{HDA}$ “adds missing cells”)

- ④ So $L * M$ is recognized by a finite HDA, hence regular.

Monadic Second-Order Logics over Ipomsets

Definition

MSO-iiPoms is built using the grammar

$$\begin{aligned} \psi ::= & a(x) \mid s(x) \mid t(x) \mid x < y \mid x \dashrightarrow y \mid x \in X \mid \\ & \exists x. \psi \mid \forall x. \psi \mid \exists X. \psi \mid \forall X. \psi \mid \psi_1 \wedge \psi_2 \mid \psi_1 \vee \psi_2 \mid \neg \psi \end{aligned}$$

- signature $\{<, \dashrightarrow, (a)_{a \in \Sigma}, s, t\}$
- $x \prec y := x < y \wedge \neg(\exists z. x < z < y)$
- for $\psi \in \text{MSO}$ with free variables $x_1, \dots, x_n, X_1, \dots, X_m$ and $P \in \text{iiPoms}$, a **valuation** is $\nu = (\nu_1, \nu_2)$ with $\nu_1: \{x_1, \dots, x_n\} \rightarrow P$ and $\nu_2: \{X_1, \dots, X_m\} \rightarrow 2^P$
- $P \models_\nu \psi$ if ψ holds with x_i and X_j interpreted as $\nu(x_i)$ and $\nu(X_j)$
- $P \models \psi$ if $P \models_{(\emptyset, \emptyset)} \psi$ (no free variables: a sentence)
- $L(\psi) = \{P \in \text{iiPoms} \mid P \models \psi\}$
- L is **MSO-definable** if $\exists \psi \in \text{MSO} : L = L(\psi)$

Example

$$\varphi = \exists x \exists y. a(x) \wedge b(y) \wedge \neg(x < y) \wedge \neg(y < x)$$

- satisfied by all P which contain an a in parallel with a b

- in particular any conlists $\begin{bmatrix} \vdots \\ a \\ \vdots \\ b \\ \vdots \end{bmatrix}$ or $\begin{bmatrix} \vdots \\ b \\ \vdots \\ a \\ \vdots \end{bmatrix}$

- but not by ab nor ba

$\Rightarrow L(\varphi)$ has **infinite** width and is **not** closed under subsumption

Results

Theorem

For all $L \subseteq \text{iiPoms}$,

- if L is *MSO-definable*, then $L_{\leq k} \downarrow$ is *regular* for all $k \in \mathbb{N}$;
(subsumption closure of width restriction of L)
- if L is *regular*, then L is *MSO-definable*.

The constructions are effective in both directions.

Corollary

For all $k \in \mathbb{N}$, a language $L \subseteq \text{iiPoms}_{\leq k}$ with $L = L \downarrow$ is *regular* iff it is *MSO-definable*.

Results

Corollary

For all $k \in \mathbb{N}$ and $\varphi \in \text{MSO}$ with $L(\varphi) = L(\varphi)_{\leq k} \downarrow$, *satisfiability* and *model checking* are *decidable*.

- also implied by Courcelle's theorem, even more generally for $L(\varphi) = L(\varphi)_{\leq k}$, because $\text{iiPoms}_{\leq k}$ has bounded treewidth

Corollary

For all $k \in \mathbb{N}$ and $L \subseteq \text{iiPoms}_{\leq k}$, if L is MSO-definable, then so is $L \downarrow$.

- does **not** hold for general (non-interval) ipomsets [Fanchon-Morin 2009]

Proofs: Special ipomsets

Definition

An ipomset $(P, <, \dashrightarrow, S, T, \lambda)$ is

- **discrete** if $<$ is empty (hence $\dashrightarrow$ is total)
- a **starter** if it is discrete and $T = P$
- a **terminator** if it is discrete and $S = P$
- an **identity** if it is both a starter and a terminator

$$\begin{bmatrix} \bullet a \\ \bullet b \bullet \\ a \bullet \end{bmatrix}$$

$$\begin{bmatrix} \bullet a \bullet \\ \bullet b \bullet \\ a \bullet \end{bmatrix}$$

$$\begin{bmatrix} \bullet a \\ \bullet b \bullet \\ \bullet a \bullet \end{bmatrix}$$

$$\begin{bmatrix} \bullet a \bullet \\ \bullet b \bullet \\ \bullet a \bullet \end{bmatrix}$$

Lemma

Any interval ipomset has a decomposition as a sequence of starters and terminators.

$$\begin{bmatrix} a \longrightarrow b \\ c \longrightarrow a \end{bmatrix} = \begin{bmatrix} a \bullet \\ c \end{bmatrix} * \begin{bmatrix} \bullet a \\ a \bullet \end{bmatrix} * \begin{bmatrix} b \\ \bullet a \end{bmatrix} = \begin{bmatrix} a \bullet \\ c \bullet \end{bmatrix} * \begin{bmatrix} \bullet a \bullet \\ \bullet c \end{bmatrix} * \begin{bmatrix} \bullet a \bullet \\ a \bullet \end{bmatrix} * \begin{bmatrix} \bullet a \\ \bullet a \bullet \end{bmatrix} * \begin{bmatrix} b \bullet \\ \bullet a \bullet \end{bmatrix} * \begin{bmatrix} \bullet b \\ \bullet a \end{bmatrix}$$

Unique decompositions

Notation: **St**: set of starters ${}_S U_U$
Te: set of terminators ${}_U U_T$
Id = **St** $\cap$ **Te**: set of identities ${}_U U_U$
 Ω = **St** $\cup$ **Te**

Definition

A word $w = (S_1, U_1, T_1) \dots (S_n, U_n, T_n) \in \Omega^+$ is **coherent** if $T_i = S_{i+1}$ for all i .
The set of coherent words is denoted **Coh**.

Definition

A coherent word is **sparse** if proper starters and proper terminators are alternating.

- additionally, all $w \in \text{Id} \subseteq \Omega^+$ are sparse
- so that's $\text{Id} \cup (\text{St} \setminus \text{Id})((\text{Te} \setminus \text{Id})(\text{St} \setminus \text{Id}))^* \cup (\text{Te} \setminus \text{Id})((\text{St} \setminus \text{Id})(\text{Te} \setminus \text{Id}))^*$

Lemma

Any interval ipomset P has a **unique** decomposition $P = P_1 * \dots * P_n$
such that $P_1 \dots P_n \in \Omega^+$ is **sparse**.

From MSO to HDAs

Idea: translate ipomsets to decompositions in Ω^+ and MSO-iiPoms to MSO over Ω

- problem: Ω is infinite $\implies$ need to restrict width

Lemma

For every $k \in \mathbb{N}$ and $\varphi \in \text{MSO}_{\text{iiPoms}}$ there exists $\hat{\varphi} \in \text{MSO}_{\Omega_{\leq k}}$ such that $P_1 \dots P_n \models \hat{\varphi}$ iff $P_1 \dots P_n$ is *coherent* and $P_1 * \dots * P_n \models \varphi$.

- $\text{coherent}_{\leq k} := \forall x \forall y. x \prec y \implies \bigvee_{P_1 P_2 \in \text{Coh}_{\leq k} \cap \Omega_{\leq k}^2} P_1(x) \wedge P_2(y)$

- ...

- Let $K = \{P \in \text{iiPoms}_{\leq k} \mid P \models \varphi\}$

$\implies L'' = \{P_1 \dots P_n \in \Omega_{\leq k}^+ \mid P_1 * \dots * P_n \in K\}$ is $\text{MSO}_{\Omega_{\leq k}}$ -definable (lemma)

- standard Büchi & Kleene $\implies L''$ is $\Omega_{\leq k}$ -rational
- replace concatenation by gluing $\implies L = \{P \in \text{iiPoms}_{\leq k} \mid P \models \varphi\} \downarrow$ is rational

From HDAs to MSO

Idea: encode accepting paths into MSO

- (similar to classical construction)
- for $P \in \text{iiPoms}$, use sparse decomposition $P = P_1 * \dots * P_n$ to define $st : P \rightarrow \{-\infty\} \cup \mathbb{N}$, $te : P \rightarrow \mathbb{N} \cup \{+\infty\}$ to denote where events start and terminate in $P_1 \dots P_n$

$\Rightarrow e \in P_i$ iff $st(e) \leq i \leq te(e)$

- and then map st and te to upsteps and downsteps in an accepting path

Lemma

For $f, g \in \{st, te\}$ and $\bowtie \in \{=, <, >\}$, the relations $f(x) \bowtie g(y)$, $\min(f)$ and $\max(f)$ are MSO-definable.

From HDAs to MSO

Lemma

Let X be an HDA and $P \in \text{iiPoms}$ with sparse decomposition $P = P_1 * \dots * P_n$. Then $P \in L(X)$ iff there exist $\rho_{\nearrow} : P \setminus S_P \rightarrow \text{ups}(X)$ and $\rho_{\searrow} : P \setminus T_P \rightarrow \text{downs}(X)$ such that for all $e_1, e_2 \in P$,

- $st(e_1) = st(e_2) \implies \rho_{\nearrow}(e_1) = \rho_{\nearrow}(e_2)$ and $te(e_1) = te(e_2) \implies \rho_{\searrow}(e_1) = \rho_{\searrow}(e_2)$
- $st(e_2) = te(e_1) + 1 \implies \text{src}(\rho_{\nearrow}(e_2)) = \text{tgt}(\rho_{\searrow}(e_1))$
- $te(e_2) = st(e_1) + 1 \implies \text{src}(\rho_{\searrow}(e_2)) = \text{tgt}(\rho_{\nearrow}(e_1))$
- $\rho_{\nearrow}(e_1) = (p, \nearrow^A, q) \implies A = \{e \mid st(e) = st(e_1)\} \ \& \ \text{ev}(q) = \{e \mid st(e) \leq st(e_1) < te(e)\}$
- $\rho_{\searrow}(e_1) = (p, \searrow_A, q) \implies A = \{e \mid te(e) = te(e_1)\} \ \& \ \text{ev}(q) = \{e \mid st(e) < te(e_1) \leq te(e)\}$
- $st(e_1) = 1 \implies \text{src}(\rho_{\nearrow}(e_1)) \in \perp_X$ and $te(e_1) = 1 \implies \text{src}(\rho_{\searrow}(e_1)) \in \perp_X$
- $st(e_1) = n \implies \text{tgt}(\rho_{\nearrow}(e_1)) \in \top_X$ and $te(e_1) = n \implies \text{tgt}(\rho_{\searrow}(e_1)) \in \top_X$

... and now **encode these into MSO**

Conclusion

- Higher-Dimensional Automata: an automaton-like model for non-interleaving concurrency
- With a **nice** language theory!
- ST-automata provide a good combinatorial abstraction
- and geometry & topology provide intuition

Ongoing & Future work:

- **Linear-time** logic for HDAs
- **Branching-time** logics for HDAs
- Higher-dimensional **timed** automata
- from **Petri nets** to HDAs
- HDAs over **infinite** ipomsets
- Active **learning** of HDAs
- **Games** on HDAs

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